AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

Chemistry Paper I

Time allowed: 35 minutes Marks 25

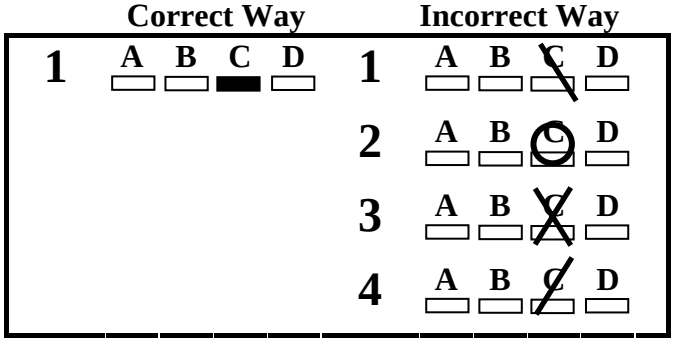
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. The relative atomic mass of nitrogen present in a molecule of ammonia is

   A. 14 amu.
   B. 17 amu.
   C. 24 amu.
   D. 28 amu.

2. Which of the following types of reaction is shown below?

   $$CuO_{(s)} + H_2SO_{4(aq)} \rightarrow CuSO_{4(aq)} + H_2O_{(l)}$$

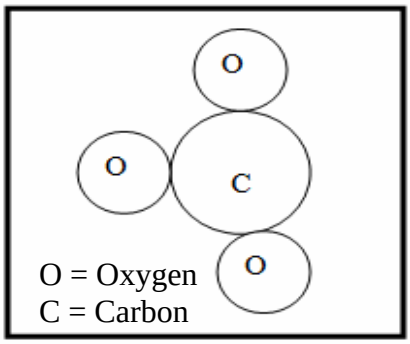
   A. Addition
   B. Combustion
   C. Decomposition
   D. Neutralisation

3. Supply the co-efficients necessary to balance the equation below.

   $$_K_{(s)} + _H_2O_{(l)} \rightarrow __KOH_{(aq)} + __H_{2(g)}$$

   A. 2:2:2:1
   B. 1:2:1:2
   C. 2:1:2:1
   D. 1:2:2:2

4. The charge on the molecular ion shown in the diagram will be

   A. $^{-}1$
   B. $^{-}2$
   C. $^{+}1$
   D. $^{+}2$

5. Which of the following configurations represents an Mg^{2+} ion ?

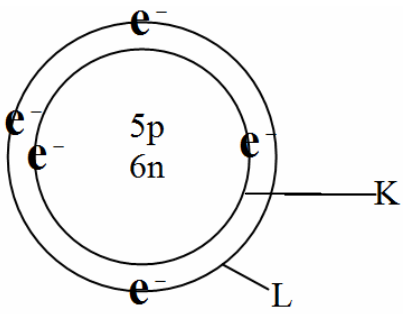
   A. $1s^2 2s^2 2p^6 3s^2$
   B. $1s^2 2s^2 2p^6 3s^1$
   C. $1s^2 2s^2 2p^6$
   D. $1s^2 2s^2 2p^2$

6. Chlorine has two isotopes which may be represented as ^{37}Cl and ^{35}Cl .

Which of the given information about ^{37}Cl is correct?

	Proton	Neutron	Electron
A	17	17	20
B	20	17	20
C	17	20	17
D	17	20	20

7. The given atomic structure is of



A. boron.
B. carbon.
C. lithium.
D. sodium.

8. Which of the following statements about the periodic table are correct?

I. Elements are arranged in order of increasing atomic mass
II. A period is a horizontal row
III. The properties of the elements within a period vary from element to element.

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

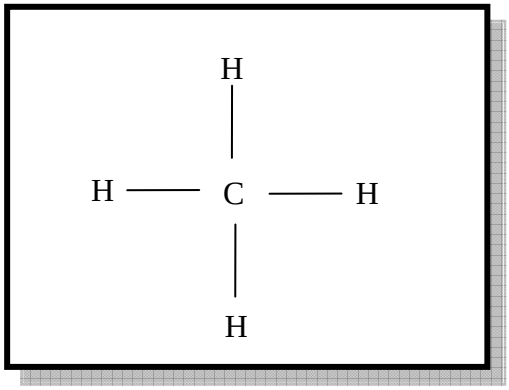
9. When we go down the group VII, the elements

A. become more reactive.
B. become bigger in size.
C. become mono atomic.
D. attain low melting and boiling points.

10. Which of the following represents the compound of a transition metal?

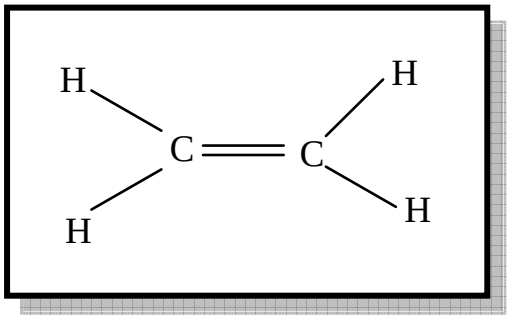
Metal	Melting point	Colour of the compound
A	High	Blue
B	Low	Green
C	High	White
D	Low	White

11. Which type of bond is shown in the given figure?



A. Ionic
B. Covalent
C. Co-ordinate covalent
D. Metallic

12. How many pi-electrons are found between C=C in the following compound?



A. 2
B. 3
C. 4
D. 5

13. Which of the following does NOT conduct electricity?

A. Molten NaOH
B. Molten KOH
C. Solid NaCl
D. Aqueous NaCl

14. Which of the following is an example of diffusion?

A. Bubbles rising in a beaker of boiling water
B. Fragrance of flowers in a garden
C. Swelling of raisins in water
D. Steam condensing on a cold window

15. When impurities is added into water, the boiling point of the water

A. increases.
B. decreases.
C. remains unchanged.
D. becomes unpredictable.

16. Which of the following is responsible for change of state in matter?

A. Density
B. Mass
C. Pressure
D. Temperature

17. If 21 cm^3 of alcohol is present in 70 cm^3 of its aqueous solution, the percentage by volume of alcohol will be

A. 21%
B. 20%
C. 30%
D. 35%

18. Which of the following information shows that the solution is supersaturated?

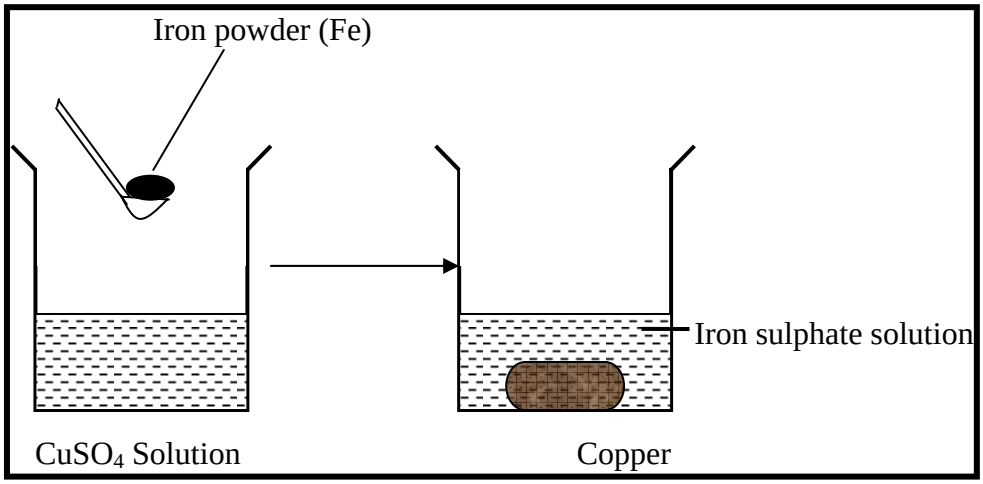
Type of solution	Compound	Temperature	Mass of solute in 100 ml of water
A	NH_3	90ºC	10g
B	NH_4Cl	80ºC	70g
C	Na_2SO_4	80ºC	20g
D	$NaNO_3$	25ºC	10g

19. Which of the following predictions is WRONG about the solubility of substance **X** in substance **Y**?

	Substance X	Substance Y	Solubility
A	Methane	Petrol	Soluble
B	NaCl	Water	Soluble
C	Grease	Water	Insoluble
D	Paint	Kerosene oil	Insoluble

20. Which substance is reduced in the given reaction?

A. Iron
B. Copper
C. Sulphur
D. Oxygen



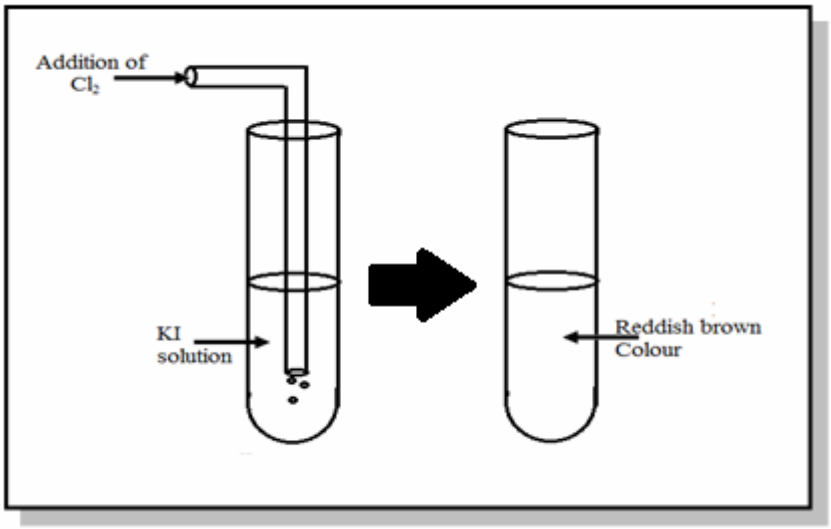
21. Which of the following is the oxidation state of phosphorus in H_3PO_4?

A. +2
B. +3
C. +4
D. +5

22. Which of the following should be added during chromium plating to prevent hydrolysis?

A. Antimonial lead
B. Chromium sulphate
C. Sulphuric acid
D. Sodium hydroxide

23. When Cl_2 is bubbled in the KI solution test tube, the solution turns reddish brown due to the liberation of



A. iodine.
B. potassium.
C. potassium chloride.
D. potassium iodide.

24. Which of the following is formed when a metal reacts with oxygen?

A. Hydroxides
B. Basic oxide
C. Acidic oxide
D. Neutral oxide

25. Which of the following reacts vigorously with dilute acids to produce hydrogen gas and salt?

A. Calcium
B. Copper
C. Silver
D. Zinc

END OF PAPER

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